wind. It is also called supporting tissues. Fibres have a great commercial value in cottage and textile industries.



Fibres are the longest plant cells. Longest Fibres occur in *Boehmeria* (Ramie fibre) 55 cm long

Fibres are of five types

1. Wood Fibres or Xylary Fibres

These fibres are associated with the secondary xylem tissue. They are also called xylary fibres. These fibres are derived from the vascular cambium. These are of two types. a. Libriform fibres b. Fibre tracheids

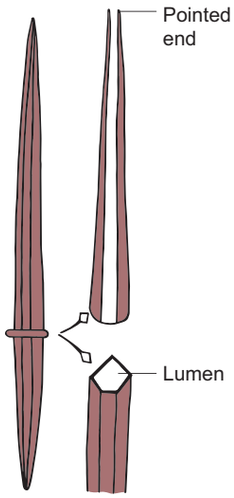


Figure 9.9 T.S of fibre

2. Bastfibres or Extra Xylary Fibres

These fibres are present in the phloem. Natural Bast fibres are strong and cellulosic. Fibres obtaining from the phloem or outer bark of jute, kenaf, flax and hemp plants. The so called pericyclic fibres are actually phloem fibres.

3. Surface Fibres

These fibres are produced from the surface of the plant organs. Cotton and silk cotton are the examples.They occur in the testa of seeds.

4. Mesocarp Fibres

Fibres obtained from the mesocarp of drupes like coconut.

5. Leaf Fibres

Fibres obtained from the leaf of *Musa, Agave and Sensciveria.*

Fibres in Our Daily Life

Economically fibres may be grouped as follows

1. **Textile Fibres:** Fibres utilized for the manufacture of fabrics, netting and cordage etc.
   a. **Surface Fibres**: Example: Cotton.
   b. **Soft Fibres**: Example: Flax, Jute and Ramie
   c. **Hard fibres:** Example: Sisal, Coconut, Pineapple, Abaca *etc.*
2. **Brush fibre:** Fibres utilized for the manufacture of brushes and brooms.
3. **Rough weaving fibres:** Fibres utilized in making baskets, chairs, mats etc.
4. **Paper making fibres:** Wood fibres utilized for paper making.
5. **Filling fibres:** Fibres used for stuffing cushions, mattresses, pillows, furniture etc. Example: *Bombax* and Silk cotton.

Complex Tissues

A complex tissue is a tissue with several types of cells but all of them function together as a single unit. It is of two types – xylem and phloem.

Xylem or Hadrome

The xylem is the principal water conducting tissue in a vascular plant. The term xylem was introduced by **Nageli**(1858) and is derived from the Gk. *Xylos* – wood. The xylem which is derived from Procambium is called **primary xylem** and the xylem which is derived from vascular cambium is called **secondary xylem**. Early formed primary xylem elements are called protoxylem, whereas the later formed primary xylem elements are called metaxylem.

Protoxylem lies towards the periphery and metaxylem that lies towards the centre is called **Exarch.** It is common in *roots.*

Protoxylem lies towards the centre and meta xylem towards the periphery this condition is called **Endarch**. It is seen in *stems.*

Protoxylem is located in the centre surrounded by the metaxylem is called **Centrarch**. In this type only one vascular strand is developed. Example: *Selaginella sp.*

Protoxylem is located in the centre surrounded by the metaxylem is called **Mesarch**.In this type several vascular strands are developed. Example: *Ophioglossum sp.*

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